# Retrieval/Matching Models

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# Sparse Term-based Retrieval

- Create inverted-index for documents
- Traverse documents linked by the inverted list of the words in query and score them by retrieval models such as BM25 and query likelihood

**Note**

When the inverted index consists of passages, the score of the document is integrated by its passage scores

# Dense Embedding-based Retrieval

- Express each document and the query as a dense vector
- Retrieve documents that are the nearest $k$ neighbors of the query vector

# Datasets

- MS MACRO Retrieval
- MS MACRO Rank

# Metrics

- Mean Reciprocal Rank (MRR)

$$\text{MRR} = \sum_{q \in \mathcal{Q}} \frac{1}{r}, \tag{1}$$

where $r$ is the rank of the first ground-truth relevant document.

- Recall@K

$$\text{Recall@K} = \frac{\#\text{ relevant item within top } K}{\#\text{ total relevant items}} \tag{2}$$

# Re-weight Methods

Re-weight the word in the document accoding to the contextualized embedding produced by Language Models. Utilize exact match and inverted index to retrieve.

# DeepCT

**Motivation:** Generate the contextualized word weight according to its embeddings (produced by BERT).

- $$y = Wx + b,$$

  where $x$ is the BERT embedding of the word, $W$ and $b$ are parameters.
- The weights is used to construct inverted index and applied in the first-stage retrieval.
- The ground-truth weight for each term in the document is estimated by the coocurrance of the term in the doc and the query, then use Mean Square Loss to train the regression model.

# HDCT

**Motivation:** Generate contextualized term importance at the document-level, exceeding the word limitation of BERT.

- weighting terms within a passage (about 300 consecutive word sequence), then aggregate passages of a document.
- document-level evidence (whether a word appears in the important document fields e.g. titles and links, and whether the word appears in the query for the document) provides labels for training HDCT, which can down-weight unimportant documents.
- similar framework as DeepCT.

# Expansion Methods

Expand the document/query to contain richer information. Utilize exact match and inverted index to retrieve.

# Pseudo-feedback

**Motivation:** Simulate the first round of user feedback, facilitating the user to get improved retrieval performance without an extended interaction.

$$q^* = q + \lambda \sum_{i=1}^{k} d_i,$$

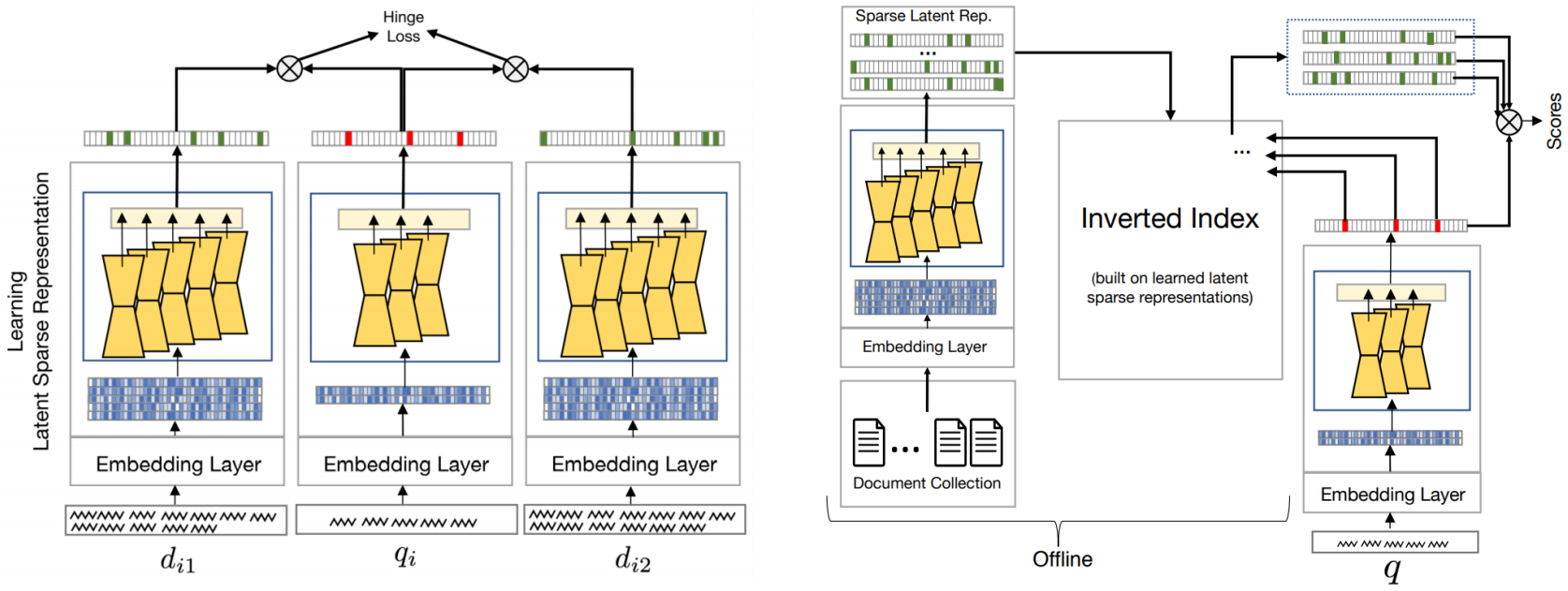
where $d_i$ is the sparse vector of the relevant document with $q$ accoding the retrieval model. $q^*$ is the finalized query vector.

# Doc2Query

**Motiation:** Generate a set of possible queries for each document using a seq-to-seq model and expand the document with these queries, in order to expand the *receptive field* of the document and facilitate retrieval models e.g. BM25.

# SNRM

**Motivation:** Create a latent representation that aims to capture meaningful semantic relations while still efficiently matching documents.




(a) Training time (b) Inference time

- The learnt representation of the document and the query should be sparse, and be in the same semantic space. The representation of the query should be sparser than that of the document, to ensure few computation when retrieval.

$$s(q, d) = \phi_Q(q)^\top \phi_D(d),$$

$$\phi_*(x) = \mathsf{ReLU}\left(\frac{1}{|d| - n + 1} \sum_{i=1}^{|d|-n+1} \phi(w_i, \cdots, w_{i+n-1})\right),$$

$$\phi(w_i, \cdots, w_{i+n-1}) = \mathsf{MLP}\left(\bigoplus_{j=i}^{i+n-1} E_{w_j}\right).$$

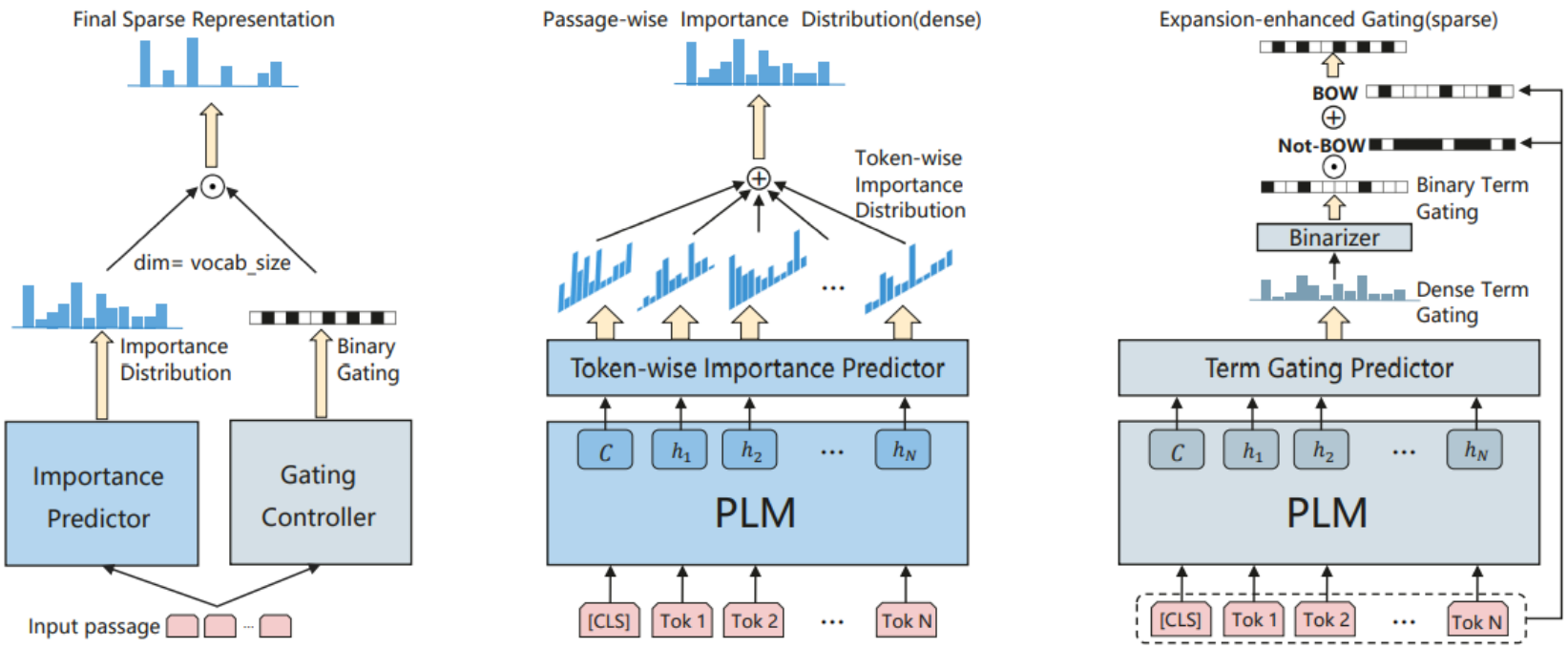
**Insight**

N-grams are used as $w_i$, I think Bert-based embeddings would be better.

- In order to train the model to be effective in retrieval and sparse in representation, a unified loss composed of a hinge-loss and a L1-norm of the representation vector are proposed. The training signal (which one is more relevant to $q$ between $d_1$ and $d_2$) is given by a pretrained query likelihood model.

# SparTerm

**Motivation:** Learn deep term-based sparse representation for each passage in the whole vocabulary space rather than the given text.




(a) SparTerm Model (b) Importance Predictor (c) Gating Controller

- Importance predictor: output the dense importance distribution of the whole passage.

$$I = \sum_{i=1}^{n} \text{transform}(r_i) E + b \in \mathbb{R}^{|V|},$$

where $r_i$ is the Bert representation of the $i$-th word in the given passage, and $E$ is the embedding matrix composed of the representation of every word, $|V|$ denotes the size of the vocabulary and transform is a function to normalize $r_i$.

- Gating Controller: output binarizer to impose the importance distribution to be sparse.

$$\hat{G} = G_{exp} + \text{BoW}(p),$$
$$G_{exp} = (G > \gamma) \odot \neg \text{BoW}(p),$$

where $\text{BoW}(p)$ is the binary BoW vector, $G$ is the gating

distribution obtained by another *importance predictor* with independent parameters. $\gamma$ is the threshold for $G$ to be included in the expansion.

- Combine the two module to produce the sparse representation (term-based) for passage:

$$s(p) = \hat{G} \odot I(p)$$

- End-to-end training, Maximum log likelihood for *Importance predictor* and Cross entropy for *gating controller*.

**Insight**

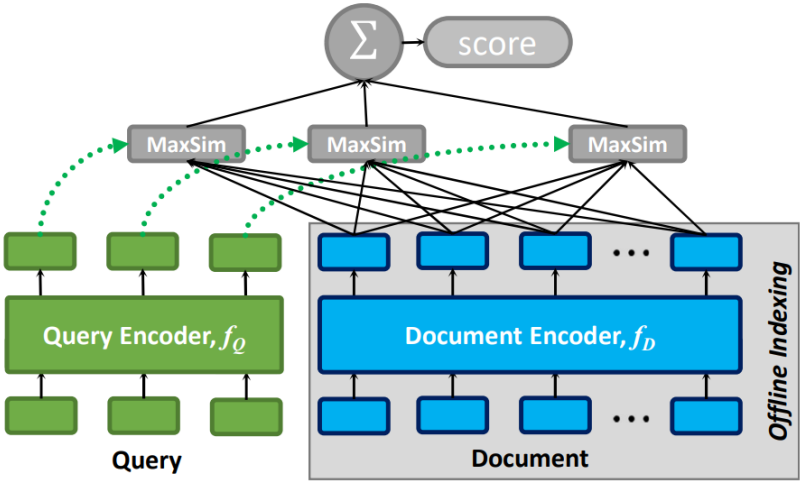
Equals to DeepCT when $G_{exp}$ is $\mathbf{0}$

# Interaction Methods

Use neural models to learn representation for each word in the document and the query, which interacts online.

# ColBert

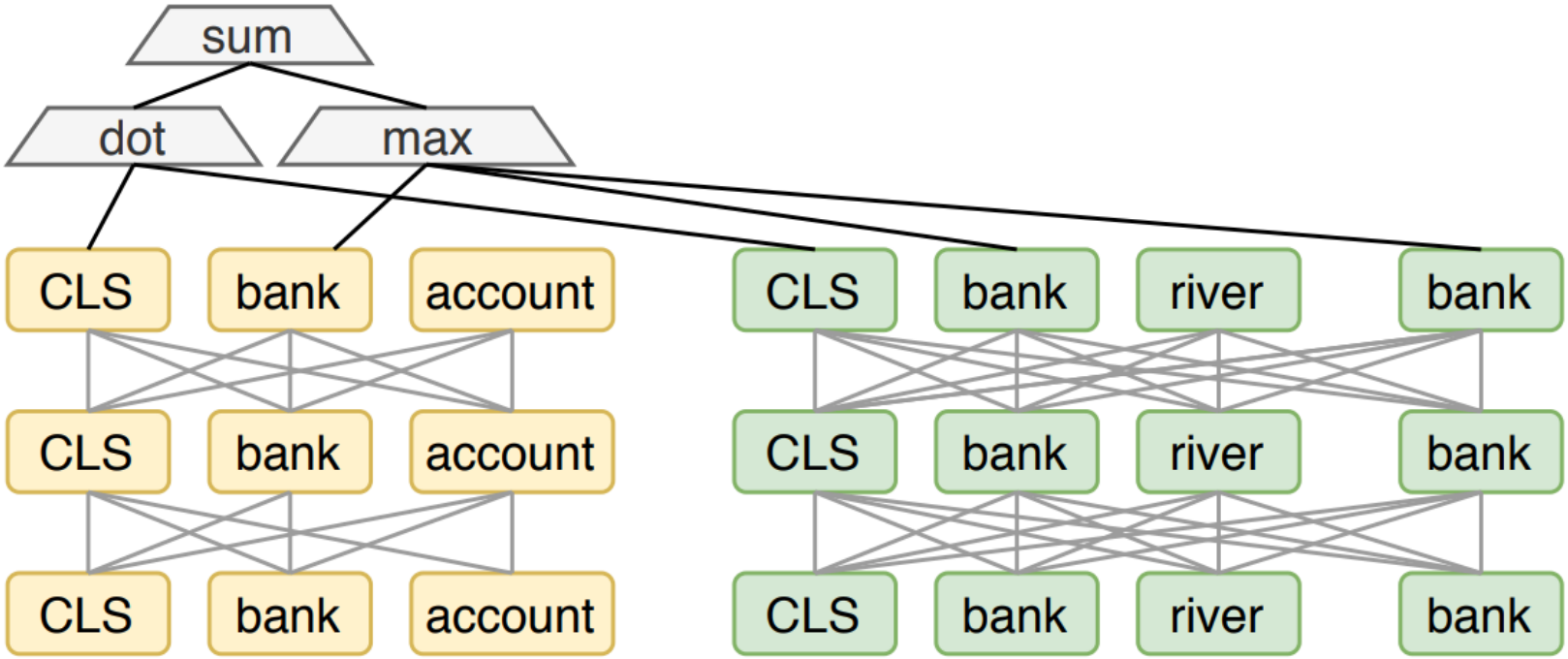
**Motivation:** Bert-based model is effective but slow, thus use *late interaction* (i.e. online interaction) to speed up inference.




- each query term interacts with each document term, the max score of the document term is kept, then the score of every query term is summed to form the final document score.
- MaxSim is compatible with approximate indexing, so can be used for ANN search (retrieval).

# COIL

**Motivation:** Performs exact query-document token matching but compute matching signals with contextualized token representations instead of heuristics.

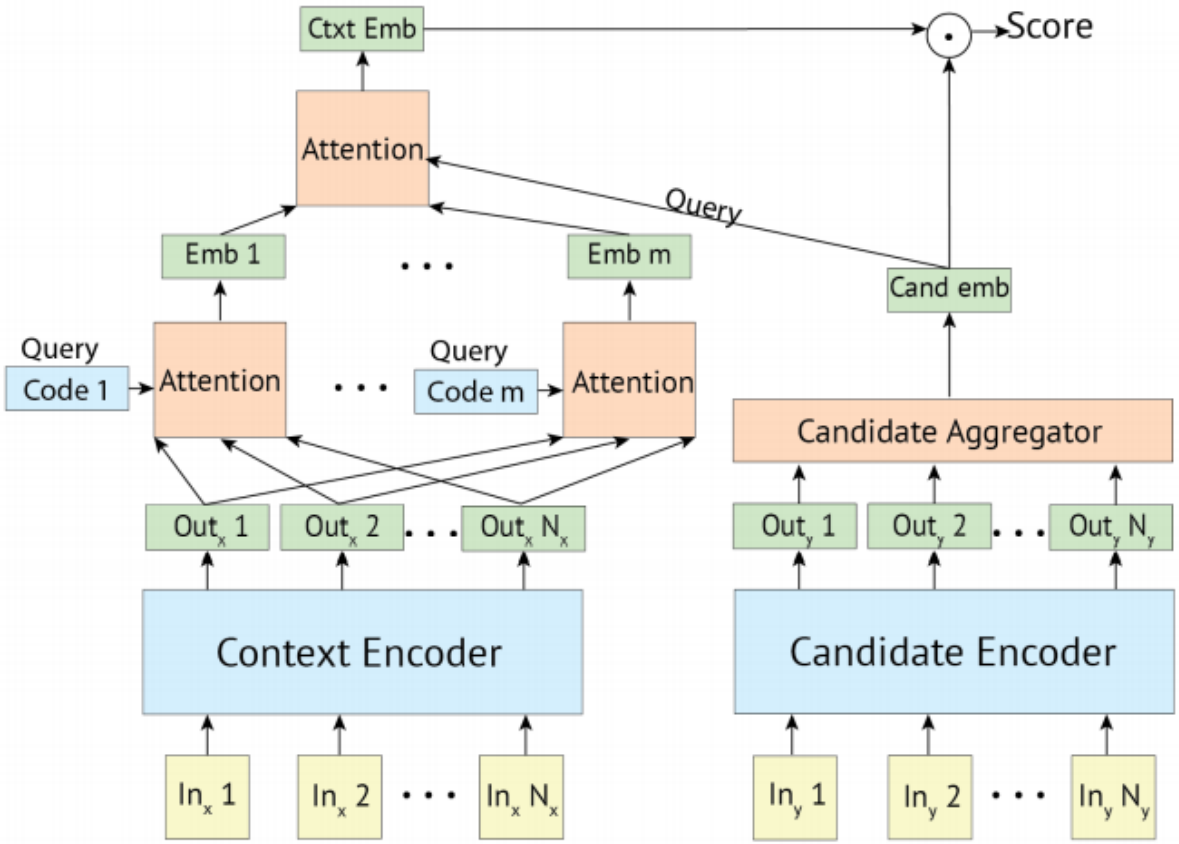


- $$s(q,d) = \sum_{t \in q \cap d} h_q(q,t)^\top h_d(d,t) + \mathsf{CLS}_q^\top \mathsf{CLS}_d$$

  where $q$ is a query, $d$ is a document and $t$ is a co-occuring term between them. $h_q(q,t)$ gives contextualized embedding for $t$ in $q$, $h_d(d,t)$ gives that of $t$ in $d$.
- The contextualized vector of the same word $h_q(q,t), h_d(d,t)$ can be stored in an inverted list to be used for retrieval. Further, the vector can be structured by approximate index to speed up searching.

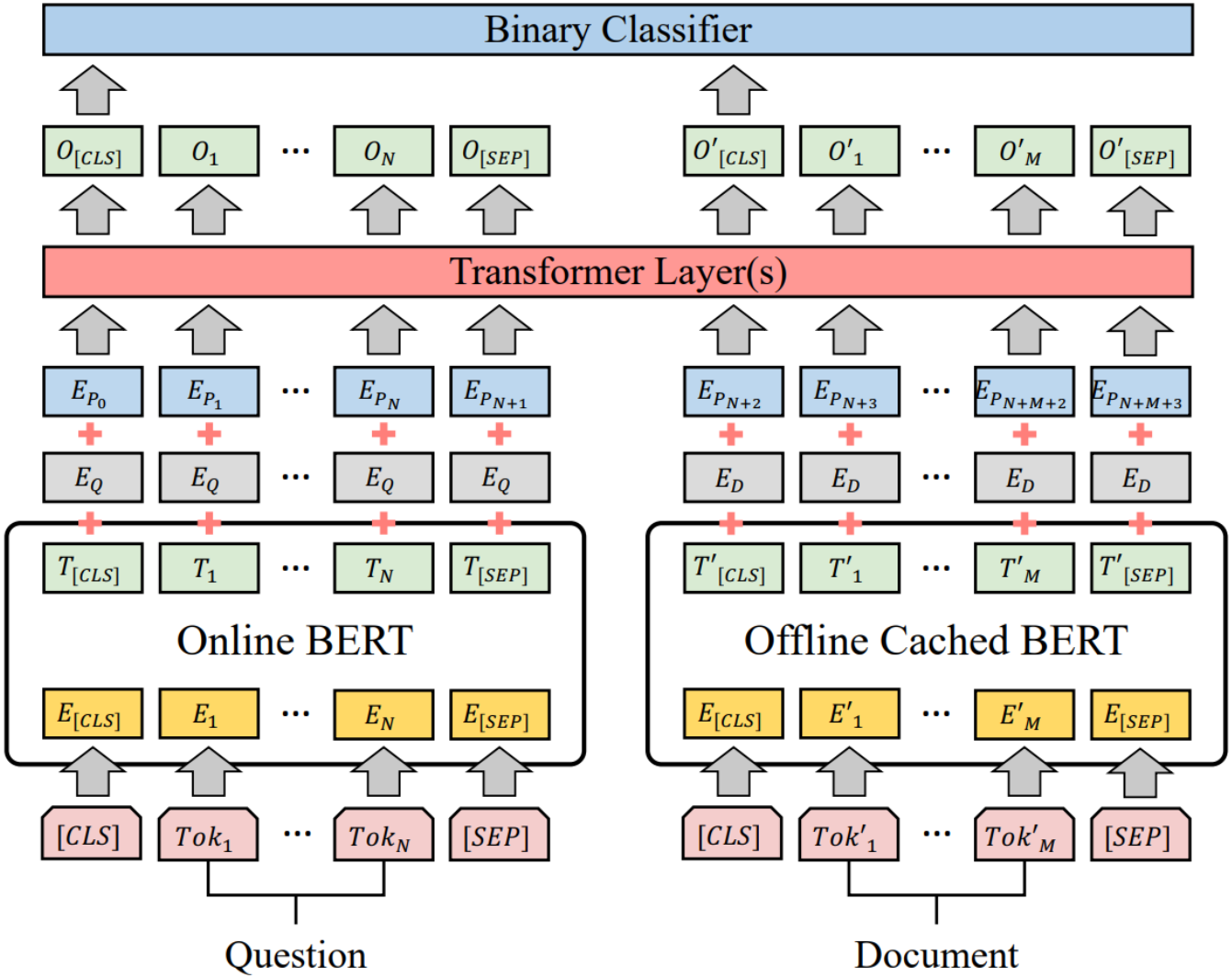
# PolyEncoder

**Motivation:** Get the best of both worlds from the Bi- and Cross-encoder, where the formmer is efficient and the latter is effective.



# DC-Bert

**Motivation:** Similar to polyencoder.



# Intra-document Cascading

**Motivation:** Ranking documents of varying length involves scoring each passage (a fixed size window) and aggregate the passage scores. However, scoring each passage through a large pre-trained model (e.g. Bert) is expensive. This work proposes to apply a light model to select only several important passages to generate the document score.

- An effective ETM model i.e. Bert is trained standalone for ranking relavant documents.
- A light ESM model is then trained in a teacher-student manner for knowledge distillation, which is taught to select the top $K$ passages with the highest score produced by ETM.

# My Insight

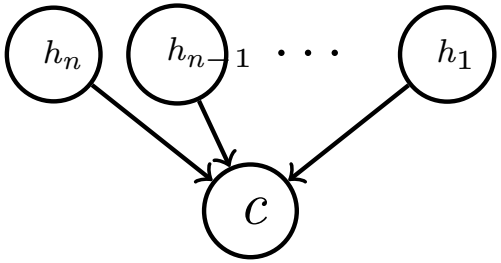
## Retrieval

- I think sparse representation for document/query is desirable because the inverted index induced is fast and the retrieval would be intepretable. However, sparse representation cannot tackle vocabulary mismatch, so maybe we can learn some dense vector named activation head (like in Octopus) for each sparsity unit, and retrieve by activating several sparsity unit according to the similarity between these activation head.
- The task of document summarization is natually related to the generation for such sparsity unit, maybe VAE can be applied to learn latent and meaningful words to represent the document.
- How about multi-task? One is document summarization and the other is retrieval, both tasks are optimized together.

# My Insight

## Selection

- Recommendation is slightly different with retrieval, where no query is provided. However, we can generate the query accoding to the history of the user.
- I prefer the query to be **bag-of-words**. In this case, the simplest way is to select key terms from each historical news as the query (like memory network). This isn't optimal since **temporal factors** are lost. Also, there may be duplicates among historical documents, therefore, each term in the query should be unique and attached with different weights.
- If the query is **a sequence of words**, I think seq-to-seq models can be utilized?

# Proposal

- Two profiles for a user: a **global** one that contains only one query, which is generated from the whole history, and a **fine-grained** one that contains multiple queries, which is generated from each of the documents.
- The global query is used to retrieve documents from the whole collection. Each retrieved document is processed to be a new query and concatenated to the end of the fine-grained profile, where we can derive a score mesuring the probability that the user clicks it by interaction models.